

SIMPLE EDA ON MARVEL UNIVERSE CHARACTERS DATASET



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KASUS

Laporan ini menggunakan dataset yang didapat dari *website* [kaggle.com](https://www.kaggle.com) yang berisi informasi terperinci tentang pahlawan super dan penjahat Marvel, termasuk nama asli, kekuatan, afiliasi, dan peran mereka dalam Marvel Universe. Kumpulan data ini menampilkan data tentang berbagai karakter ikonik dari Marvel Comics, seperti Iron Man, Captain America, Thor, dan banyak lainnya. Berikut dataset karakter Marvel yang digunakan:

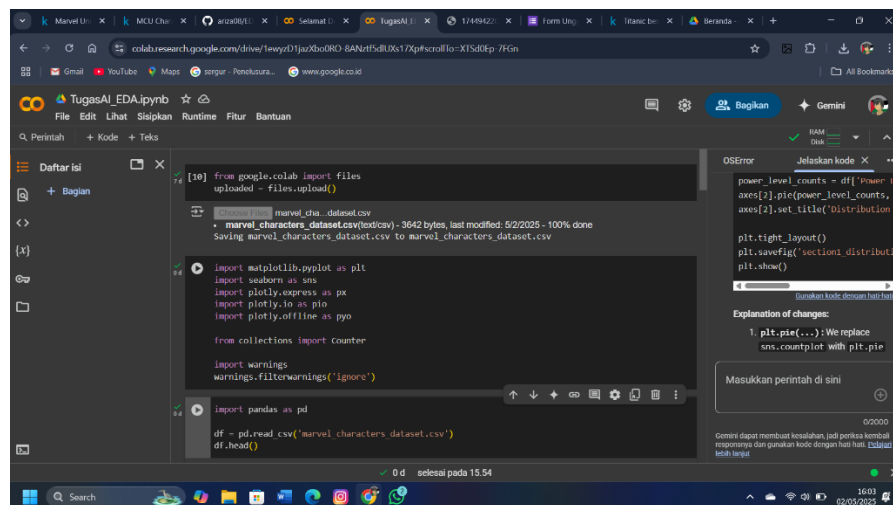
Character	Real Name	Affiliation	Powers	Role	Power Level
Iron Man	Tony Stark	Avengers	Powered Armor, Genius-level intellect	Hero	Low
Captain America	Steve Rogers	Avengers	Super Soldier, Enhanced strength	Hero	Low
Thor	Thor Odinson	Avengers	God of Thunder, Weather manipulation	Hero	Low
Black Widow	Natasha Romanoff	Avengers	Superhuman strength, Espionage	Hero	Low
Hulk	Bruce Banner	Avengers	Superhuman strength, Gamma Radiation	Hero	Low
Spider-Man	Peter Parker	Avengers	Spider sense, Wall crawling	Hero	Low
Wolverine	James Howlett	X-Men	Regeneration, Adamantium claws	Hero	Low
Deadpool	Wade Wilson	X-Force	Regeneration, Skilled hand-to-hand combatant	Antihero	Low
Black Panther	T'Challa	Avengers	Superhuman strength, Vibranium suit	Hero	Low
Doctor Strange	Stephen Strange	Avengers	Sorcery, Magic	Hero	Low
Ant-Man	Scott Lang	Avengers	Size manipulation, Ant communication	Hero	Low
Scarlet Witch	Wanda Maximoff	Avengers	Telekinesis, Reality warping	Hero	Low
Vision	Victor Von Doom	Avengers	Genius intellect, Magic	Hero	Low
Daredevil	Matt Murdock	Defenders	Enhanced senses, Superhuman strength	Hero	Low
Luke Cage	Carl Lucas	Defenders	Superhuman strength, Bulletproof skin	Hero	Low
Gamora	Gamora Zenwhoberi	Guardians of the Galaxy	Superhuman strength, Expert fighter	Hero	Low
Star-Lord	Peter Quill	Guardians of the Galaxy	Weapons expert, Green energy manipulation	Hero	Low
Rocket Raccoon	Rocket	Guardians of the Galaxy	Cybernetic enhancements	Hero	Low
Black Widow	Natasha Romanoff	Avengers	Rocket science, Genius intellect	Hero	Low
Thanos	Thanos	Villain	Expert marksmanship, Advanced technology	Villain	Low
Doctor Doom	Victor Von Doom	Villain	Superhuman strength, Regeneration	Villain	Low
Green Goblin	Norman Osborn	Villain	Magical powers	Villain	Low

Loki	Loki Laufeyson	Avengers	Superhuman speed, Enhanced senses	Hero	Low
Hawkeye	Clint Barton	Avengers	Aerial combat, Shield mastery	Hero	Low
Bucky Barnes	James Buchanan Barnes	Avengers	Superhuman strength, Regeneration	Hero	Low
Falcon	Sam Wilson	Avengers	Chi manipulation, Martial arts expert	Hero	Low
Shang-Chi	Shang-Chi	Avengers	Skilled combatant, Weapon mastery	Hero	Low
Moon Knight	Marc Spector	Avengers	Energy manipulation, Expert marksman	Hero	Low
Elektra	Elektra Natchios	Defenders	Skilled combatant, Enhanced reflexes	Hero	Low
Mantis	Mantis	Avengers	Superhuman strength, Agility	Hero	Low
Valkyrie	Brunnhilde	Defenders	Superhuman intelligence, Tentacle control	Hero	Low
Black Bolt	Blackagar Boltagon	Avengers	Illusion creation, Hypnosis	Hero	Low
Silver Surfer	Norrin Radd	Inhumans	Superhuman strength, Durability	Hero	Low
Red Skull	Johann Schmidt	Fantastic Four	Symbiote bonding, Enhanced strength	Villain	Low
Ultron	Henry Pym	Hydra	Flight, Superhuman strength, Cosmic energy	Villain	Low
Hela	Sue Storm	Avengers	Magical powers	Villain	Low
Mister Fantastic	Johnny Storm	Fantastic Four	Elasticity, Superhuman strength	Hero	Low
Invisible Woman	Ben Grimm	Fantastic Four	Invisibility, Force field generation	Hero	Low
Human Torch	Otto Octavius	Fantastic Four	Pyrokinesis, Flight	Hero	Low
Thing	Quentin Beck	Sinister Six	Superhuman strength, Durability	Hero	Low
Doctor Octopus	Mac Gargan	Sinister Six	Superhuman strength, Tentacle control	Hero	Low
Mysterio	Eddie Brock	Sinister Six	Illusion creation, Hypnosis	Villain	Low
Rhino	Carol Danvers	Sinister Six	Superhuman strength, Durability	Villain	Low
Venom	Eddie Brock	Avengers	Symbiote bonding, Enhanced strength	Antihero	Low
Captain Marvel	Carol Danvers	Avengers	Flight, Superhuman strength, Cosmic energy	Hero	Low

ANALISIS

Analisis dataset dalam laporan ini menggunakan *website* Google Colab untuk prosesnya. Berikut link dan codenya:

<https://colab.research.google.com/drive/1ewyzD1jazXbo0RO-8ANztft5dlUXs17Xp#scrollTo=ZDFGVie9DKIF>



```
[10] from google.colab import files
      uploaded = files.upload()

      _____ marvel_characters_dataset.csv
      • marvel_characters_dataset.csv (text/csv) - 3642 bytes, last modified: 5/2/2025 - 100% done
      Saving marvel_characters_dataset.csv to marvel_characters_dataset.csv

0.0 import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.io as pio
import plotly.offline as pyo

from collections import Counter

import warnings
warnings.filterwarnings('ignore')

0.0 import pandas as pd

df = pd.read_csv('marvel_characters_dataset.csv')
df.head()
```

OSError

```
power_level_counts = df['power_level_counts']
axes[2].pie(power_level_counts,
axes[2].set_title('Distribution')

plt.tight_layout()
plt.savefig('section1_distribution.png')
plt.show()
```

Explanation of changes:

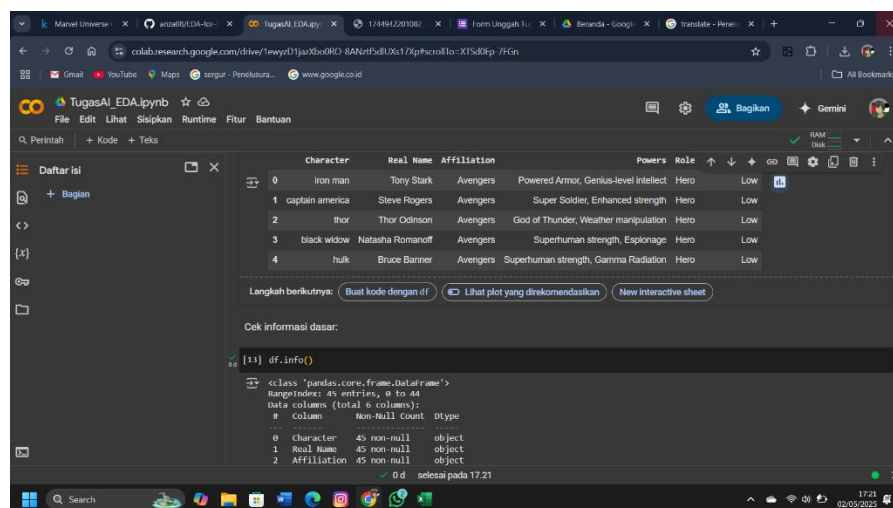
1. `plt.pie(...)`: We replace `sns.countplot` with `plt.pie`

Masukkan perintah di sini

0.0000

Gemini dapat membuat kesalahan, jadi periksa kembali responsnya dan gunakan kode dengan hati-hati. [Laporkan](#) [Batal](#)

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```
0.0 df.head()
```

	Character	Real Name	Affiliation	Powers	Role	
0	iron man	Tony Stark	Avengers	Powered Armor, Genius-level intellect	Hero	Low
1	captain america	Steve Rogers	Avengers	Super Soldier, Enhanced strength	Hero	Low
2	thor	Thor Odinson	Avengers	God of Thunder, Weather manipulation	Hero	Low
3	black widow	Natasha Romanoff	Avengers	Superhuman strength, Espionage	Hero	Low
4	hulk	Bruce Banner	Avengers	Superhuman strength, Gamma Radiation	Hero	Low

Langkah berikutnya: [Buat kode dengan df](#) [Lihat plot yang direkomendasikan](#) [New interactive sheet](#)

Cek informasi dasar:

```
[13] df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45 entries, 0 to 44
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   Character   45 non-null    object
 1   Real Name   45 non-null    object
 2   Affiliation 45 non-null    object
```

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```
colab.research.google.com/drive/1ewyD1ja2Xb0RO-8ANt5dUX617Xp#scrollTo=XTSd0Ep-7FGn

TugasAI_EDA.ipynb

File Edit Lihat Sisipkan Runtime Fitur Bantuan

Perintah + Kode + Teks

Daftar isi + Bagian

[x]

df.duplicated().sum()

rp.int64(0)

Ringkasan statistik untuk beberapa kategori:

print("\nvalue counts for Role:")
print(df['role'].value_counts())
print("\nvalue counts for Affiliation:")
print(df['Affiliation'].value_counts())
print("\nvalue counts for Power Level:")
print(df['Power Level'].value_counts())

Value Counts for Role:
Role
Hero      35
Villain    8
Antihero   7
Name: count, dtype: int64

Value Counts for Affiliation:
Affiliation
Avengers    23
Defenders    4
```

```
colab.research.google.com/drive/1ewyD1ja2Xb0RO-8ANt5dUX617Xp#scrollTo=XTSd0Ep-7FGn

TugasAI_EDA.ipynb

File Edit Lihat Sisipkan Runtime Fitur Bantuan

Perintah + Kode + Teks

Daftar isi + Bagian

[x]

df.describe()

Character      Real Name  Affiliation      Powers  Role  Power Level
count         45         45         45         45  45         45
unique         44         41         10         38   3         1
top    black widow  Natasha Romanoff  Avengers  Superhuman strength, Durability  Hero    Low
freq           2         2         23         3   35         45

Cek value null dan duplikat:

df.isnull().sum()

Character      0
Real Name      0
Affiliation     0
Powers         0
```

```
colab.research.google.com/drive/1ewyD1ja2Xb0RO-8ANt5dUX617Xp#scrollTo=XTSd0Ep-7FGn

TugasAI_EDA.ipynb

File Edit Lihat Sisipkan Runtime Fitur Bantuan

Perintah + Kode + Teks

Daftar isi + Bagian

[x]

Memvisualisasikan dataset:

sns.set_theme(style='darkgrid')
sns.set_palette('deep')

# Visualisasi distribusi menggunakan pie chart
fig, axes = plt.subplots(1, 3, figsize=(10, 5))

# Distribusi role
role_counts = df['role'].value_counts()
axes[0].pie(role_counts, labels=role_counts.index, autopct='%1.1f%%', startangle=90)
axes[0].set_title('Distribution of Character Roles')

# Distribusi affiliation
affiliation_counts = df['Affiliation'].value_counts()
axes[1].pie(affiliation_counts, labels=affiliation_counts.index, autopct='%1.1f%%', startangle=90)
axes[1].set_title('Distribution of Affiliations')

# Distribusi power level
power_level_counts = df['Power Level'].value_counts()
axes[2].pie(power_level_counts, labels=power_level_counts.index, autopct='%1.1f%%', startangle=90)
axes[2].set_title('Distribution of Power Levels')

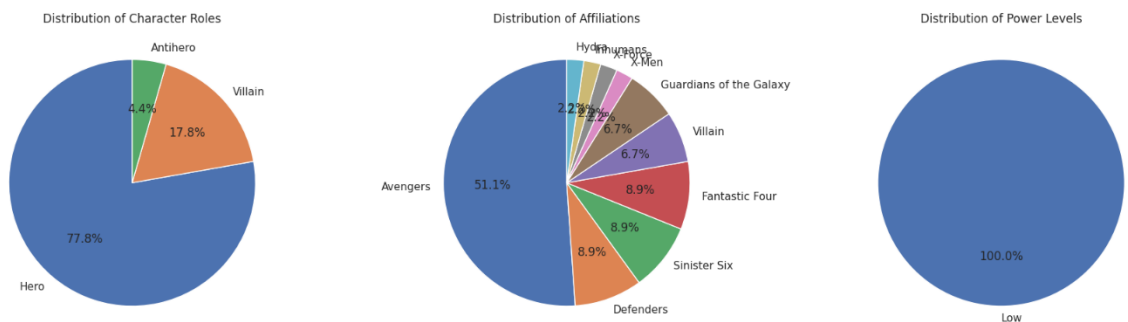
plt.tight_layout()
plt.show()
```

HASIL

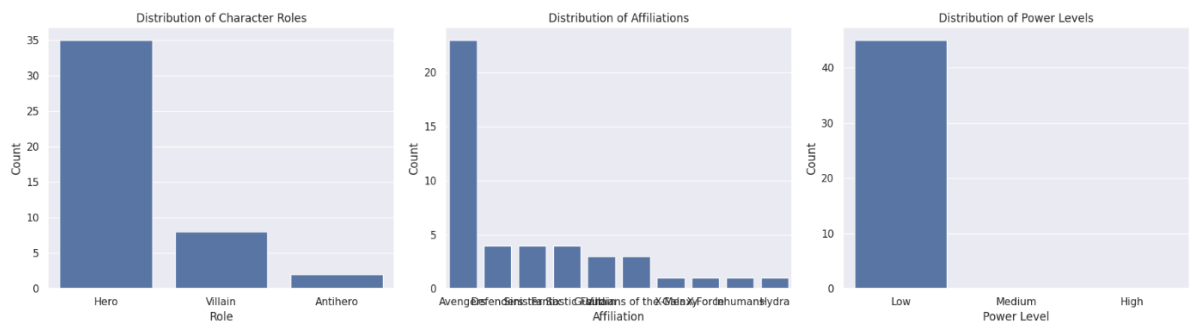
Hasil dari EDA dari dataset *Marvel Characters Universe* adalah sebagai berikut:

A. Visualisasi

Berikut hasil visualisasi jika menggunakan Pie chart:



Berikut hasil visualisasi jika menggunakan Diagram:



Kemudian setelah semua proses EDA selesai, file EDA dari Google Colab disimpan dalam Github. Berikut link-nya: <https://github.com/ariza08/EDA-for-MCU-dataset>

KESIMPULAN

Laporan ini menggunakan dataset yang didapat dari *website* [kaggle.com](https://www.kaggle.com) yang berisi informasi terperinci tentang pahlawan super dan penjahat Marvel, termasuk nama asli, kekuatan, afiliasi, dan peran mereka dalam Marvel Universe. Kumpulan data ini menampilkan data tentang berbagai karakter ikonik dari Marvel Comics, seperti Iron Man, Captain America, Thor, dan banyak lainnya. Lalu analisis dataset dalam laporan ini menggunakan *website* Google Colab untuk prosesnya. Hasil dari EDA dari dataset *Marvel Characters Universe* adalah menggunakan Pie Chart dan Diagram.